

CO, AT<sub>3</sub>

**SIMATS ENGINEERING  
ASSESSMENT TOOL 3**

**COURSE NAME:** Artificial Intelligence

**COURSE CODE:** CSA1714

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**COURSE OUTCOME COVERED**

**CO1:** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

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**QUESTIONS:** 2

**Duration :** 45 Minutes  
**Total Marks:** 20

Annexure A

**Descriptive Written Test**

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**Code of Conduct:**

I SRINIVASAN.A / 192A11025 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:**

A. Sriniv

Level - 3.

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## CSAITH - Artificial Intelligence

### Smart home automation Systems.

A Smart home automation control Systems controls lighting, temperature, and security using sensors and intelligent agents. Different types of intelligent agent perform different tasks.

#### 1. Simple Reflex agent.

- \* works using condition action rules
- \* Takes decisions only from the current sensor input.

Ex:

- \* If motion is detected  $\rightarrow$  Turn on lights
- \* If no motion for 10 minutes  $\rightarrow$  Turn off lights.

- \* Advantage: Fast, simple and easy to implement
- \* Disadvantage: cannot remember past events.

#### 2. Model Based Reflex Agent.

- \* Maintains an internal model of home environment
- \* Uses current and previous sensor information.

Ex:

- \* Remembers that the bedroom light is already ON.
- \* Detects if a window is open while the AC is running and turns off the AC.



\* Advantages : Handles partially observable environments and makes better decisions.

\* Disadvantages : More complex and requires memory.

### 3. Goal Based Agent

\* Makes decisions to achieve specific goals

Ex :

\* Maintain room temperature at  $24^{\circ}\text{C}$

\* Keep the home secure.

Advantages : Flexible and can plan future actions.

Disadvantages : Requires planning, which increases computation time.

### 4. Utility Based Agent

\* Choose the action that provides the highest overall benefit

\* Balances comfort, energy efficiency, and security

Ex :

Slightly reduces AC usage during peak electricity hours to save energy while maintaining comfort.

Advantages : Produces optimal decisions in complex situations.

Disadvantages : Utility calculations can be computationally expensive.



# Smart Traffic Management System:

A Smart Traffic management system controls traffic lights using vehicle density, pedestrian movement, weather conditions and historical traffic data.

## 1. Simple Reflex Agent;

Responds only to current traffic conditions.

Ex:

If vehicle count is high  $\rightarrow$  Extend green signal.

Advantages: Fast response

Disadvantages: Cannot predict future traffic.

## 2. Model Based Agent;

\* Maintains a model of current traffic conditions.

Ex:

Identifies congestion building up and adjusts nearby signals.

Advantages: Better traffic coordination

Disadvantages: Requires more processing and memory.

## 3. Goal Based Agent;

\* works to achieve goals such as:

\* Reduce congestion

\* Minimize waiting time

\* Improve pedestrian safety.

\* Plans signal timings to achieve these goals.



Advantages: Flexible and intelligent.

Disadvantages: Planning may take additional computation.

#### 4. Utility-Based Agent:

\* Evaluates multiple factors such as travel time, fuel consumption, emergency vehicle priority, and pedestrian safety.

Advantages: Produces the best overall traffic flow.

Disadvantages: Utility calculations are complex.

#### Most Effective Approach:

A hybrid system combining Model-Based and Utility-Based Agents is the most effective solution. The model based agent maintains an accurate view of traffic conditions, while the utility based agent optimizes signal timing by balancing congestion, travel time, emergency vehicle movement, and pedestrian safety. This combination provides efficient, adaptive and intelligent traffic management.